Labs

Alveary

About the Course

This is the required lab component for level 6 General Science. This course includes our custom laboratory book which will build foundational and progressive skills and habits.

This topic is included in the following course(s): Science: Grade 6

About Science: Grade 6
In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level and time required are transitional as we gently prepare for the middle grades.

Science: Grade 6
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more "friends" in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge "by the way" and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 6
In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level, historical context, and time required are transitional as we gently prepare for the middle grades.

Labs: Grade 6
The ideas and skills in this component are progressive, like math or grammar. Teachers should read the lab book thoroughly to understand what concepts might need to be supplemented should they choose a different sequence or a substitution.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
6	Labs: Grade 6 1 time/week 30 min	Science: Grade 6 Lab Book

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 6				
Natural History: Grade 6		General Science: Grade 6 Nature Walks & Scouting: Grades 1-8	General Science: Grade 6 Nature Notebook: Grade 6	Labs: Grade 6

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 6
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Labs: Grade 6
☐ Preview science labs and purchase materials, or have students gather them. Print or shortcut any links, as desired. While it does not usually matter when the week's lab is done in reference to the week's readings, this is not the case for Art of Construction in Term 3. The lab instructions for this book are embedded in the text, and the text is most living when they are completed alongside the reading. Therefore, it is best for lab day to fall on the third lesson of the week. If students' lab day is earlier in the week, teachers may want to delay the start of week 1, so that the third lesson is lab day. For example, if students have M/Th lesson days and T lab day, then students could begin lessons on Th of week 1. Alternatively, teachers may look ahead and adjust according to their preference.

Special Topics & Field Trips

Science: Grade 6
Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

Term Prep & Teacher Tips

Science: Grade 6
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

- View Book List Details

Labs: Grade 6

Science: Grade 6 Lab Book

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

- View Basic Supplies
- View Supply List Details

Labs: Grade 6


Hand Lens


Scale/Balance


Classroom Collection of Rocks and Minerals


General Fossil Collection

Quick Links

Labs: Grade 6

- Grade 6 Lab Book
- Lab Notebook Examples

Click THIS text or scan the QR code for links.

Labs: Grade 6

How To Teach

Introduce

Regardless of how many days are required to complete a particular activity, every science lab has the same flow, which follows the scientific method and is guided by the lab book.

• On day 1, learners read an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is analogous to the conversation we might have when we begin something new in any subject, but they may need to dialogue as they learn to extend these skills to the laboratory.

• Once they have had a chance to think, they will compose a prelab narration to put these introductory ideas into their notebooks. The prompt in the lab is generalized and consistent, so they learn the habit. Eventually, they will learn to formulate this as a hypothesis. For example, a learner preparing for a lab about the use of insect repellent might write:

"I have read about some diseases that are spread by insects, like Lyme disease. I also know that my sister is allergic to some insect bites. Insect repellent contains pesticides to keep insects away. Some scientists worry about how pesticides affect wildlife. I am going to compare some different insect repellants in this lab to see if they really work."

• Written narration and composition are skills that they will build over time. These prelab narrations may seem short and even incomplete at first, but that is okay. If learners have difficulty or are easily frustrated, then provide them with support. Teachers may act as scribes or allow students to keep a digital notebook to type or use assistive technology, as appropriate.

• After they complete their prelab narration, the listed materials are collected. This gives the learner some responsibility to let the teacher know if something is missing or to remind the teacher if something needs to be purchased at the store.

• If these activities on day 1 do not fill the scheduled time, that is fine. They might use the additional time to familiarize themselves with the procedure, draw a picture from their book, or catch up on any other work. Some labs may instruct the student to begin on the same day.

Lab Procedure

• Then, students begin and follow the procedure (whenever prompted by the lab book). Note that the lesson plans guide teachers as to which sections are completed each week, and the lab book instructs learners when to take a break.

• The lab gives instructions for using their notebooks to create tables and figures, as needed. Do not allow this to become an obstacle. Do it with them, first having them watch and then having them copy or help when ready.

• If teachers choose to have learners record directly in the lab book or on a photocopy, then cut out and tape these into their lab books, so that they can see how the record is built.

• Learners may feel unsatisfied with their results. This is often part of the process. Come alongside, help them learn to be okay with uncertainty and questions, and teach them what to do with it in the next step.

Analysis & Conclusions

• The last step in the lab is to analyze the data and observations and draw conclusions from them. For some simpler labs, learners will complete their analysis and concluding (or postlab) narration on the last day of the lab procedure. For labs that are more involved, a separate day has been built in to allow adequate time for this.

• Similar to the prelab narration, the concluding narration is a chance to think about and put into words (or questions). This time, they are considering what they learned from the lab, what they could learn more about if they were to continue, and possibly how they would pursue that learning. Again, they might need support in the form of dialogue, a scribe, etc. For example, the above learner might write:

“It was clear that the Off and black pepper essential oil worked against ants because they would not even touch the line of repellent, but I wasn’t sure about the Skin So Soft because it ran all over the place. I could test this one again with different insects or on a different surface.”

● Depending on the interest of the learner and the priorities of the teacher, the student might be encouraged to spend more time on those ideas of what more they could learn, or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.

● Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

Labs: Grade 6

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 1

⬚ **30m Labs: Grade 6 - Lesson 1**
Lab: Reading Rocks

▯ Materials: Grade 6 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of Reading Rocks, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in the lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary. Spend remaining time gathering materials for next week.

★ TEACHER TIP
Pacing is only a suggestion. Students should engage with the lab at a pace that is appropriate for their abilities and interests.

WEEK 2

⬚ **30m Labs: Grade 6 - Lesson 2**
Lab: Reading Rocks

▯ Materials: Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Reading Rocks, as directed in the Lab Book.

Suggested day 2: Students complete roughly steps 1-3 of the Procedure today. There is some flexibility to complete more or less, depending on the student.

WEEK 3

⬚ **30m Labs: Grade 6 - Lesson 3**
Lab: Reading Rocks

▯ Materials: Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of Reading Rocks, as directed in the Lab Book.

Suggested day 3: Students complete steps 4-8 of the Procedure today.

• NATURE NOTEBOOK
Note that step 8 of lab directs students to conduct outdoor work before the next lab period: see prompt in Outdoor Work Quick Link.

WEEK 4

⬚ **30m Labs: Grade 6 - Lesson 4**
Lab: Reading Rocks

▯ Materials: Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 4 of Reading Rocks, as directed in the Lab Book.

Suggested day 4: Students complete step 9 of the Procedure today, as well as the Analysis and Conclusion section.

WEEK 5

⬚ **30m Labs: Grade 6 - Lesson 5**
Lab: Fossils

▯ Materials: Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete Fossils, as directed in the Lab Book.

Labs: Grade 6

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 6

☐ **30m Labs: Grade 6 - Lesson 6**
Lab: What Do Volcanoes Tell Us?

Materials: Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of What Do Volcanoes Tell Us?, as directed in the Lab Book.

Suggested day 1: Students read the Introduction and then compose the prelab narration in the lab notebook. Use any remaining time to begin the Procedure.

WEEK 7

☐ **30m Labs: Grade 6 - Lesson 7**
Lab: What Do Volcanoes Tell Us?

Materials: Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of What Do Volcanoes Tell Us?, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure today.

WEEK 8

☐ **30m Labs: Grade 6 - Lesson 8**
Lab: What Do Volcanoes Tell Us?

Materials: Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of What Do Volcanoes Tell Us?, as directed in the Lab Book.

Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book.

WEEK 9

☐ **30m Labs: Grade 6 - Lesson 9**
Lab: Earthquake Shake

Materials: Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of Earthquake Shake, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and compose the prelab narration in the lab notebook. Use any remaining time to begin the Procedure.

WEEK 10

☐ **30m Labs: Grade 6 - Lesson 10**
Lab: Earthquake Shake

Materials: Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Earthquake Shake, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure today.

Labs: Grade 6

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 11	⬚ **30m Labs: Grade 6 - Lesson 11** *Lab: Earthquake Shake* Materials: Grade 6 Lab Book and materials listed within **→ LAB DAY** Complete day 3 of Earthquake Shake, as directed in the Lab Book. Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book.	
WEEK 12	⬚ **30m Labs: Grade 6 - Lesson 12** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: Grade 6 Lab Book	

Labs: Grade 6

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 1	⬚ **30m Labs: Grade 6 - Lesson 13** *Lab: Classification* Materials: Grade 6 Lab Book and materials listed within PREP: Read Teacher Tip **➔ LAB DAY** Complete day 1 of Classification, as directed in the Lab Book. Suggested day 1: Students read the Introduction and then compose the prelab narration in the lab notebook. These need not be more than 1-3 sentences at first, and teachers should feel free to scribe for students, as necessary.	★ TEACHER TIP Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interests. ★ TEACHER NOTE Lab 1 introduces the idea of "relatedness" between species. Teachers might choose to allow the idea to pass by the way, to consider how they would like to discuss, or what other theories they might like to present.
WEEK 2	⬚ **30m Labs: Grade 6 - Lesson 14** *Lab: Classification* Materials: Grade 6 Lab Book and materials listed within **➔ LAB DAY** Complete day 2 of Classification this week, as directed in the Lab Book. Suggested day 2: Students work on the Procedure today. This is a long procedure, so a lesson period has been provided for additional time.	
WEEK 3	⬚ **30m Labs: Grade 6 - Lesson 15** *Lab: Classification* Materials: Grade 6 Lab Book and materials listed within PREP: Read Teacher Tip **➔ LAB DAY** Complete day 3 of Classification, as directed in the Lab Book. Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book.	★ TEACHER TIP Students will need fresh, unblemished apples for the next lab!
WEEK 4	⬚ **30m Labs: Grade 6 - Lesson 16** *Lab: Modeling the Skin* Materials: Grade 6 Lab Book and materials listed within PREP: Read Teacher Tip **➔ LAB DAY** Complete day 1 of Modeling the Skin, as directed in the Lab Book. Suggested day 1: Students complete the Introduction. Try to complete step 1 of the Procedure today.	★ TEACHER TIP If needed, be sure to pour your agar plates according to the package instructions at the appropriate time. They are needed in step 2 of the Procedure. ∞ **Video Link:** Pouring Agar Plates (2:51)
WEEK 5	⬚ **30m Labs: Grade 6 - Lesson 17** *Lab: Modeling the Skin* Materials: Grade 6 Lab Book and materials listed within **➔ LAB DAY** Complete day 2 of Modeling the Skin, as directed in the Lab Book. Suggested day 2: Students complete the Procedure today.	

Labs: Grade 6

[Click THIS text or scan the QR code for links.]

Term 2

WEEK 6

⬚ **30m Labs: Grade 6 - Lesson 18**
Lab: Modeling the Skin

▯ Materials: Grade 6 Lab Book and materials listed within

➔ LAB DAY
Complete day 3 of Modeling the Skin, as directed in the Lab Book.

Suggested day 3: Students complete the Analysis and Conclusion instruction in the lab book.

WEEK 7

⬚ **30m Labs: Grade 6 - Lesson 19**
Lab: Modeling the Heart

▯ Materials: Grade 6 Lab Book and materials listed within

➔ LAB DAY
Complete day 1 of Modeling the Heart, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction.

WEEK 8

⬚ **30m Labs: Grade 6 - Lesson 20**
Lab: Modeling the Heart

▯ Materials: Grade 6 Lab Book and materials listed within

➔ LAB DAY
Complete day 2 of Modeling the Heart, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure and the Analysis and Conclusions today.

WEEK 9

⬚ **30m Labs: Grade 6 - Lesson 21**
Lab: Student Design

▯ Materials: Grade 6 Lab Book and materials listed within, video and image links

PREP: Read Teacher Tip

➔ LAB DAY
Complete day 1 of Student Design, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. They should communicate their plan and any needed supplies to their teacher.

∞ Video Link: Anatomy T-shirt (3:55)
∞ Image Link: Anatomy Cake

★ TEACHER TIP
Teachers should verify that students have what they need to begin their model next week.

WEEK 10

⬚ **30m Labs: Grade 6 - Lesson 22**
Lab: Student Design

▯ Materials: Grade 6 Lab Book and materials listed within

➔ LAB DAY
Complete day 2 of Student Design, as directed in the Lab Book.

Suggested day 2: Students will work on their model for the remainder of the term, completing the Analysis and Conclusions when finished. To allow them plenty of time, a lesson period has been provided for additional time.

Labs: Grade 6

Click THIS text or scan the QR code for links.

Term 2

WEEK 11	⬚ **30m Labs: Grade 6 - Lesson 23** *Lab: Student Design* Materials: Grade 6 Lab Book and materials listed within **➜ LAB DAY** Complete day 3 of Student Design, as directed in the Lab Book. Suggested day 3: Students complete their model and finish their Analysis and Conclusions.	
WEEK 12	⬚ **30m Labs: Grade 6 - Lesson 24** *Exam Week* **➜ EXAMS** Answer question(s) related to course. Questions will come from: Grade 6 Lab Book	

Labs: Grade 6

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 1

⬚ **30m Labs: Grade 6 - Lesson 25**
Lab: Building a Tent

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ **LAB DAY**
Complete day 2 of Building a Tent, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure.

★ TEACHER TIP
Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interests.

★ TEACHER TIP
Suggested day 1 for the lab Building a Tent was scheduled in Lesson 50, so that students would be prepared for a full lesson of activity on lab day. If students have not yet completed Lesson 50, then teachers may want to complete day 1 together before beginning the lab activity.

WEEK 2

⬚ **30m Labs: Grade 6 - Lesson 26**
Lab: Tension and Compression

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

➔ **LAB DAY**
Complete Tension and Compression, as directed in the Lab Book.

WEEK 3

⬚ **30m Labs: Grade 6 - Lesson 27**
Lab: A Steel Frame... Made Out of Paper

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

➔ **LAB DAY**
Complete day 1 of A Steel Frame... Made Out of Paper, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and step 1 of the Procedure.

WEEK 4

⬚ **30m Labs: Grade 6 - Lesson 28**
Lab: A Steel Frame... Made Out of Paper

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

➔ **LAB DAY**
Complete day 2 of A Steel Frame... Made Out of Paper, as directed in the Lab Book.

Suggested day 2: Students continue the Procedure. Help them learn to plan ahead, allowing glue to dry between sessions.

WEEK 5

⬚ **30m Labs: Grade 6 - Lesson 29**
Lab: A Steel Frame... Made Out of Paper

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

➔ **LAB DAY**
Complete day 3 of A Steel Frame... Made Out of Paper, as directed in the

[Click THIS text or scan the QR code for links.]

Term 3

Lab Book.

Suggested day 3: Students continue the Procedure.

WEEK 6

☐ **30m Labs: Grade 6 - Lesson 30**
Lab: A Steel Frame… Made Out of Paper

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 4 of A Steel Frame… Made Out of Paper, as directed in the Lab Book.

Suggested day 4: Students continue the Procedure.

WEEK 7

☐ **30m Labs: Grade 6 - Lesson 31**
Lab: A Steel Frame… Made Out of Paper

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 5 of A Steel Frame… Made Out of Paper, as directed in the Lab Book.

Suggested day 5: Students continue the Procedure, if needed, and the Analysis and Conclusions.

WEEK 8

☐ **30m Labs: Grade 6 - Lesson 32**
Lab: Strings and Sticks

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 1 of Strings and Sticks, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and steps 1-2 of the Procedure today.

WEEK 9

☐ **30m Labs: Grade 6 - Lesson 33**
Lab: Strings and Sticks

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Strings and Sticks, as directed in the Lab Book.

Suggested day 2: Students continue with the Procedure.

WEEK 10

☐ **30m Labs: Grade 6 - Lesson 34**
Lab: Strings and Sticks

▯ Materials: Art of Construction, Grade 6 Lab Book and materials listed within

→ LAB DAY
Complete day 3 of Strings and Sticks, as directed in the Lab Book.

Suggested day 3: Students continue with the Procedure.

Labs: Grade 6

[Click THIS text or scan the QR code for links.]

Term 3

WEEK 11	⬚ **30m Labs: Grade 6 - Lesson 35** *Lab: Strings and Sticks* Materials: Art of Construction, Grade 6 Lab Book and materials listed within **➔ LAB DAY** Complete day 4 of Strings and Sticks, as directed in the Lab Book. Suggested day 4: Students complete the Procedure and the Analysis and Conclusions.
WEEK 12	⬚ **30m Labs: Grade 6 - Lesson 36** *Exam Week* **➔ EXAMS** Answer question(s) related to course. Questions will come from: Art of Construction